Peer learning for our institute

Student Tutor App
Every student can ask. Every student can teach.

A campus platform where doubts reach the right peer, sessions happen on campus , and every helper builds a trusted record.

Team

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The problem

Doubts wait while knowledge sits idle

The learner

Asks a friend, a group chat, or
nobody . Hard topics pile up until
exams arrive .

The senior

Has mastered a topic but has no
easy , trusted way to share it with
those who need it .

The institute

Cannot see which topics students
struggle with , so help arrives late
or in the wrong place .

The gap : there is no trusted, organised way to connect a doubt with the right peer at the
right time .

STU DENT TUTOR APP 02

Our solution

Ask, match, meet, rate, grow

01
Ask
Post a doubt with

tags and the

outcome you expect .

02
Match

Tags route it only to
tutors who chose

that pro skill .

03
Meet
A tutor books a

campus or online
session and chats in

the app .

04

R ate

The student
confirms the session

and rates it privately .

05
Grow

Scores update

weekly , and learners
can become tutors .

Skill -tag matching
Doubts reach only tutors with that
pro skill .

Sessions inside the college

Campus spots, or a link shared in the
session chat .

Private, weekly ratings
Tutors never see who rated what .

In one line: a student asks, the right peer answers, and both grow.

STU DENT TUTOR APP 03

R oles

One account , two modes

Student mode

Ask doubts and track them
Keep personal resource buckets
Plan work in the to-do list
Add skills to the profile

Tutor mode

R eceive doubts that match your pro skills

Book and run sessions
Build a proficiency score
Edit mastered skills anytime

Kind of skill Where it lives What it does

Skills (student) Student profile Information only; never used for matching

Pro skills (tutor) Tutor profile, set on first visit Tag matching: only doubts with these tags

Why : a role depends on time and interest, so there is no role to choose at sign-up. A switch at the top right changes
mode instantly .

STU DENT TUTOR APP 04

Access · Sign up and log in

Verified students only

What it does

Sign up with name , email, institute email and roll
number , then confirm an OTP and set a password.
Log in with institute email and password.

R esults attach by roll number from the

admin’s CSV.

Why this design

Institute email plus roll number keeps the
community to real students .
No role question : Student and Tutor are modes,
not account types .

Student Tutor App

1 Details 2 OTP 3 Password

Sign up

Name

Email

Institute email

R oll number

Send OTP

Log in

Institute email

Password

Log in

Forgot password?

Wireframe · sign up and log in

STU DENT TUTOR APP 05

Access · First visit

A short, guided first visit

What it does

Student home opens with a tutorial : add skills,
add a photo , tour the pages.
Tutor mode asks for pro skills on the first visit ,
before anything else .
Pro skills come from a standard skill list.

Why this design

Matching depends on pro skills, so tutors cannot
skip that step .
Students are never pushed to become tutors .

Student Tutor App

Welcome tutorial

1 Add your skills

2 Add a photo

3 Tour the pages

Next

Choose your pro skills

Search skills

Python DBMS Calculus

Operating Systems

Electronics

Confirm skills

Wireframe · student tutorial and tutor skill picker

STU DENT TUTOR APP 06

Pages · Student home

A home that shows what needs you

What it does

Welcome note , pending doubts with time left, and
today’s plans .
A “Manage your resources” block closes
the page .
A collapsible left menu links to every
student page .

Why this design

One landing page answers “what needs me

today ?” at a glance.
Every page is one click away, so nothing hides in
sub -menus.

Student Tutor App Student Tutor

Homepage

Profile

Doubts

R esources

Leaderboard

To -do list

Good evening, Vamsi

Pending doubts
Pointers in C
18h left
Eigenvalues
2d left

Today’s plans
OS assignment

DBMS revision
Mock test

Manage your resources
Sort PDFs and links into buckets.

Wireframe · student home (sample data)

STU DENT TUTOR APP 07

Pages · Tutor home

Work first , standing second

What it does

Welcome note , doubts matched to your pro skills,
and the tutor leaderboard .
Your own position is pinned below the top three .

The menu mirrors the student side and adds

Sessions booked .

Why this design

A tutor lands on work to do before anything else.

R ank adds motivation without revealing

individual ratings .

Student Tutor App Student Tutor

Homepage
Doubts to solve

Sessions booked

Leaderboard
Analytics
Profile

Welcome back , Vamsi

Doubts for you
Binary trees
Asha · DSA

Normal forms

Karan · DBMS

Tutor leaderboard

1 Meera
2 Dev
3 Ishaan
You : #7

Wireframe · tutor home (sample data)

STU DENT TUTOR APP 08

Pages · Student profile

A profile with a target built in

What it does

Name , roll number, year, branch and programme;
phone and photo are optional .
CPI and CGPA come from the admin’s CSV and
are visible only to the student .
A target CGPA shows the average CPI needed
from this semester .

Why this design

R esults are pulled, not typed, so they stay
accurate and private .
A target turns a vague goal into one number
per semester .

Profile Student

Vamsi Adithya
25BCS270
Year 4 · CSE · B.Tech
Add phone (optional)

CPI by semester CGPA 8.4

S 1 S2 S3 S4 S5 S6

Target CGPA 8.8
Keep an average CPI of 9.1 from this semester.

Python SQL + Add skill

Wireframe · student profile (sample data)

STU DENT TUTOR APP 09

Pages · Tutor profile

R eputation, shown as totals

What it does

Shows the proficiency score , doubts solved and
experience , all system-calculated.
Mastered skills are the pro skills, editable at
any time .
Phone and photo are optional, as on the
student side .

Why this design

Only totals are visible ; single ratings and
feedback stay hidden .
Editable skills let a tutor grow into new topics.

Profile Tutor

Vamsi Adithya
25BCS270
Year 4 · CSE
Add phone (optional)

78 Proficiency score

Updated weekly

24
Doubts solved

5 mo
Experience

Mastered skills

Python DBMS + Add skill

Wireframe · tutor profile (sample data)

STU DENT TUTOR APP 10

Pages · Student doubts

Every doubt, tracked to the end

What it does

Lists your doubts, searchable by tag, in Solved
and Unsolved tabs .

Unsolved doubts show time left and let you

extend the expiry .
Solved doubts open the tutor , session time and
shared resources .

Why this design

Time left makes urgency visible ; an expired doubt
simply closes .
Everything about a doubt stays in one place after
the session .

Your doubts + Ask a new doubt

Search by tag Unsolved Solved

Pointers vs references
#C++ · 18h left

Edit expiry

Eigenvalues intuition
#Maths · 2d left

Edit expiry

Normal forms in DBMS
Solved · Meera · Fri 4 PM

Open

Wireframe · student doubts (sample data)

STU DENT TUTOR APP 11

Pages · Ask a new doubt

Know who can help before you post

What it does

Describe the doubt, choose tags, and state the
outcome you expect .
Each tag shows its tutor count, average score and
top score .
Posting reaches only tutors who chose
those tags .

Why this design

Tutor counts set expectations , so nobody posts
into an empty room .

The expected outcome tells a tutor what a good

session looks like .

Ask a new doubt

Enter your doubt

Why does my linked list lose nodes after delete ?

Choose tags

#DSA · 12 tutors · avg 72 · max 91

#C++ · 9 tutors · avg 68 · max 88

#Python · 15 tutors · avg 70 · max 94

Expected outcome

I can fix it and explain why it broke.

Cancel Post doubt

Wireframe · ask pop -up (sample data)

STU DENT TUTOR APP 12

Pages · Tutor doubts

An inbox of doubts you can answer

What it does

Lists active doubts matched to your pro skills,
with the student’s name and a summary .

Opening a doubt shows its full text and

expected outcome .
A Solved Doubts section keeps every past
session and its details .

Why this design

You never see your own doubts , so you cannot
answer yourself .
A booked doubt vanishes from other inboxes, so
there is no double booking .

Doubts to solve Tutor

Active Solved doubts

Asha · Binary trees
#DSA · 1d left

Book session

Karan · Normal forms
#DBMS · 2d left

Book session

Neha · Gradient descent
#ML · 3d left

Book session

Wireframe · tutor doubts (sample data)

STU DENT TUTOR APP 13

Pages · Book a session

Booking that locks the doubt

What it does

Asks for date, time, place, and online or offline.
Offline sessions need a campus venue ; online
ones use a link shared in the session chat .
States the cancellation rule before you confirm .

Why this design

Booking locks the doubt at once, so only one
tutor takes it .
Fixing time and venue early lets the student plan.

Book a session

Date: Fri 24 Oct Time: 4:00 PM

Mode Offline Online

Venue : Library discussion room

Cancellation rule
You cannot cancel without a replacement . If something urgent comes up,
explain it to the student .

Back Confirm booking

Wireframe · booking pop -up (sample data)

STU DENT TUTOR APP 14

Sessions · Lifecycle

From doubt to rated session

01
Doubt posted

Visible only to tutors with a matching

pro skill .

02

Session booked
The doubt leaves every other

tutor’s inbox .

03

Session chat

The tutor shares the link or venue , notes
and files .

04

Session completed

The tutor marks it done ; a form goes to
the student .

05

Student form

Did it happen? Then a private rating
and feedback .

06

Weekly update

Scores refresh weekly ; the tutor never
sees the rating .

If the tutor cannot attend
Find a replacement directly, with their consent, or broadcast a request. If nobody is found, explain to the student, who
decides on any negative rating .

STU DENT TUTOR APP 15

Sessions · Proficiency score

How the proficiency score is built

Starting weights

R ating (smoothed average) 55%

Doubts solved 20%

Doubt-solving activity 15%

Shared -resource activity 10%

R ules that keep it fair

New tutors start near the average , so one top rating cannot win the board .
R ecent sessions count slightly more than old ones.
Doubts solved is capped, so volume alone cannot buy rank.
Per-tag scores feed the average and top score
students see .

Note : the weights are starting values to tune with real data from the trial.

STU DENT TUTOR APP 16

Sessions · Trust

Keeping the leaderboard honest

R isk Safeguard

Friends trade top ratings Cap sessions per student–tutor pair; count distinct students
Sessions that never happened The student must confirm it happened , or it counts for nothing
Sudden rating spikes Auto-flag bursts, identical ratings and loops between two accounts
Unfair or revenge ratings R atings stay hidden and update weekly; a report button flags abuse
Leaderboard cheating The institute reviews top profiles and removes anyone who cannot explain

Final backstop
High-rated profiles are checked by the institute. Anyone who looks suspicious is called to the office to explain, and is
removed from the leaderboard if the answer does not hold .

Open for brainstorming : what other attacks could break this? Each one gets its own safeguard.

STU DENT TUTOR APP 17

Community · Your resources

Buckets that sort themselves

What it does

Two tabs : Your resources and Shared resources.
Create named buckets, such as Data Science,
with sub -buckets.
Each bucket sorts files into PDFs, links, photos
and text .

Why this design

Auto-sorted spaces stop a bucket becoming a
dump of mixed files .
Personal buckets need no approval, so organising
stays fast .

Your resources Shared resources

Data Science

DSA

Electronics

+ New bucket

Data Science

PDFs
6

Links

9

Photos
3

Text notes

4

+ Add a resource

Wireframe · your resources (sample data)

STU DENT TUTOR APP 18

Community · Shared resources

Shared resources , approved by the batch

01

Someone adds a file or a

new bucket .

02
“ Accept this change?”

appears for their batch and

department .

03

Voting stays open for

one day .

04
The majority decides : it goes
live or is dropped .

R ules

Only the same batch and department can see and vote . The contributor’s own vote does not count .
Proposed: a minimum vote count, otherwise the poll stays open another day .

Why : peers on the same syllabus judge relevance best.

Accept this change?
Add DBMS_PYQ_2024.pdf to PYQs › DBMS

21h left

34 accepted · 6 rejected

Accept Reject

Wireframe · vote card (sample data)

STU DENT TUTOR APP 19

Insight · Analytics and leaderboard

One dashboard for the whole campus

What it does

Shared by students and tutors : trends, solve rates
and the leaderboard .
Charts cover doubts by month, by tag and by year
of study .
Award points reward solved doubts and
high ratings .

Why this design

It shows doubts, never individual grades, so
privacy holds .
Groups under ten students are hidden to
prevent identification .

Analytics and leaderboard

412
Doubts

78%

Solved

63

Tutors

Doubts by month

Jun Jul Aug Sep Oct

Hardest tags
DBMS
Calculus

OS

Top tutors 1 Meera 92 2 Dev 88 3 Ishaan 85

Wireframe · analytics (sample data)

STU DENT TUTOR APP 20

Insight · What the data answers

What teachers and students learn

Question Answered by How it helps

Where do students struggle ? Doubts per tag and year Teachers focus revision

Which topics lack help ? Unsolved share per tag Spot topics without tutors

Is tutor supply enough? Tutors versus doubts per tag Recruit where it is thin

When is demand highest ? Monthly doubt trend Plan help before exams

Who deserves recognition ? Award points Reward consistent helpers

Privacy rule

Analytics use doubts and tags only. No individual result appears anywhere, and any group under ten students is hidden.

Award points: earned for solved doubts and high ratings, weighted so quality counts more than volume.

STU DENT TUTOR APP 21

Planner · To-do list

A planner that asks for help

What it does

Plan tasks with a date, time and priority.
Flag a task as tough when the schedule feels
too heavy .
Tough tasks show an Ask Suggestion button.

Why this design

Overloaded students need a workflow , not a full
tutoring session .
Planning and asking sit together, so help is one
tap away .

My plans + Add task

OS assignment 3
Done · Thu · High

DBMS unit 3 revision
Fri · Tough

Ask Suggestion

Mock test prep
Sat · Medium

1 of 3 done this week

Wireframe · to -do list (sample data)

STU DENT TUTOR APP 22

Planner · Ask Suggestion

A lighter kind of help

1

The student flags a task as tough and taps

Ask Suggestion.

2

A post is prefilled with the task, deadline and
time available .

3

It appears in Doubts as a Suggestion, matched to
tutors by tag .

4

Tutors reply with a short workflow and resources in a

few messages .

5

The student picks a plan ; its steps drop into the to-do
list as subtasks .

Suggestion · #DBMS

DBMS unit 3 revision · due Fri · 3 hours free. How should I plan it?

Meera’s plan
1 Skim normal forms, 40 min
2 Solve five past questions
3 Message me what is unclear

Use this plan

Wireframe · suggestion thread (sample data)

STU DENT TUTOR APP 23

Build · Security and safety

Security , privacy and safety

Identity

Institute email, roll number and OTP
verification at sign -up.

R esults

Loaded from the admin’s CSV by roll
number ; each student reads only
their own .

Database-level access

The database itself enforces who can

read what , not just the app.

Safe meetings

Sessions on campus or by a link in the

session chat ; report and block built in.

Abuse limits

R ate limits on OTPs and posts; file type
and size checks on uploads .

Minimal data

Phone and photo optional; passwords
hashed ; analytics never name students.

STU DENT TUTOR APP 24

Build · Proposed tech stack

A stack a student team can run free

Frontend

R eact + Tailwind CSS

Fast to build and responsive, with a
large component ecosystem .

Backend

Node .js + Express

R EST APIs for doubts, sessions, ratings
and votes .

Database

PostgreSQL on Supabase

R elational integrity and row-
level security .

Auth and email

Supabase Auth, email OTP

Verified sign -up without building login
from scratch .

R ealtime and files

R ealtime and Storage

Session chat , notifications, PDFs
and photos .

Hosting and jobs

Vercel or R ender

Free-tier hosting, plus scheduled jobs
for expiry and weekly scores .

Proposed, not final: chosen so a student team can build and run it without cost for a 500-user pilot.

STU DENT TUTOR APP 25

Build · Reliability

Built to stay correct and fast

One booking , one winner

Booking is a single atomic update on an open doubt, so two
tutors cannot claim it .

Tag -indexed matching

Pro skills and doubt tags live in indexed tables, so a tutor’s inbox is one fast query .

Scheduled jobs

An hourly sweep closes expired doubts; a weekly job releases
ratings and recomputes scores .

Notifications

In-app and email alerts for new matches, bookings, reminders
and the completion form .

Core tables

users pro_skills doubts sessions ratings buckets resources votes tasks

STU DENT TUTOR APP 26

In closing

What the app becomes for our college

A study group that never closes Any doubt, at any hour, answered by a peer who has been there.

A trusted tutor directory Pro skills, weekly scores and honest ratings in one place.

A live map of where students struggle Doubt trends that show teachers where to help, without

naming anyone .

A library built by the batch Resources organised, voted on and kept by the people who use them.

Every student can ask. Every student can teach.

STU DENT TUTOR APP 27